GRAVEATLAS

Phase 15 — Global Archive Federation & Future-Proof Preservation

GRAVEATLAS — PHASE 15
GLOBAL ARCHIVE FEDERATION & FUTURE-PROOF PRESERVATION

PHASE 1 THROUGH PHASE 14 MUST ALREADY BE COMPLETE.

Continue from the existing GraveAtlas project.

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CORE RULES
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- Do NOT rebuild Phases 1–14.
- Do NOT start Phase 16.
- Reuse existing architecture, security, provenance, moderation, audit, preservation, APIs, datasets, automation, intelligence and release systems.
- No paid services or external integrations without explicit approval.
- Prefer existing/free/local infrastructure where practical.
- Never expose secrets, credentials, tokens or private data.
- Never fabricate historical facts, records, institutions, users, analytics, coverage, performance, archive nodes or federation members.
- Never silently alter authoritative historical records.
- Preserve provenance and source attribution.
- Do not perform destructive or mass operations without authorization.
- Complete Phase 15 only, then stop.

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PART 1 — PHASE 14 BASELINE
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Read the actual Phase 14 implementation and final report.

Identify:
- completed capabilities
- remaining blockers
- actual digital-twin maturity
- knowledge-graph maturity
- predictive-preservation maturity
- AI limitations
- security/privacy constraints
- portability limitations
- disaster-recovery gaps

Use evidence only.

==
PART 2 — FEDERATED ARCHIVE ARCHITECTURE
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Evaluate or design a federation model supporting:
- independent archival nodes
- regional preservation nodes
- institutional archives
- trusted replicas
- synchronization
- failure recovery

Do not claim federation exists unless actually implemented.

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PART 3 — ARCHIVE NODE MODEL
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Where justified define:
- node identity
- node ownership/operator
- trust status
- dataset scope
- preservation responsibility
- synchronization state
- last verification
- capabilities

Do not expose private infrastructure information publicly.

==
PART 4 — CROSS-ARCHIVE INTEROPERABILITY
==

Support where justified:
- shared identifiers

- dataset manifests
- schema mapping
- provenance exchange
- version synchronization
- compatibility checks

Never silently discard provenance during synchronization.

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PART 5 — ARCHIVE INDEPENDENCE
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Evaluate whether GraveAtlas can operate independently of:
- one cloud provider
- one database
- one storage provider
- one API provider
- one AI provider
- one repository host

Document migration paths.

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PART 6 — PORTABLE ARCHIVAL PACKAGE
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Where practical define a portable package containing:
- data
- metadata
- identifiers
- provenance
- schema
- licensing information
- version information
- validation information
- documentation

The package must be understandable outside the current application.

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PART 7 — CRYPTOGRAPHIC INTEGRITY
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Evaluate safe mechanisms for:
- snapshot integrity
- dataset hashes
- manifest verification
- tamper detection
- signed manifests where appropriate
- chain-of-custody records

Never claim cryptographic verification without actually performing it.

Do not store private keys in source code.

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PART 8 — LONG-TERM FORMAT STRATEGY
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Identify:
- current archival formats
- obsolete/deprecated formats
- migration risks
- preferred preservation formats
- validation procedures
- migration history

Never migrate authoritative data without validation and recovery planning.

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PART 9 — FEDERATED PROVENANCE
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Preserve provenance when data moves between:
SOURCE
→ INSTITUTION
→ ARCHIVE NODE
→ GRAVEATLAS
→ PUBLIC RELEASE

Track:
- source ownership
- provenance
- transfer
- version
- timestamp

- authorization

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PART 10 — CONFLICT RESOLUTION
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Handle conflicting:
- cemetery information
- historical records
- geographic information
- source interpretations
- institutional claims

Never erase conflicting evidence merely to produce one answer.

Represent where appropriate:
- source A
- source B
- conflict
- evidence
- review status
- resolution
- reviewer

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PART 11 — GLOBAL SYNCHRONIZATION
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Where justified support:
- incremental synchronization
- version-aware updates
- integrity checks
- conflict detection
- retry limits
- failure recovery
- offline synchronization where practical

Never create uncontrolled synchronization loops.

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PART 12 — DISASTER INDEPENDENCE
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Model recovery from:
- provider failure
- regional outage
- database corruption
- storage loss
- repository loss
- API failure
- compromised replica

Test recovery safely where possible.

Never claim disaster independence without evidence.

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PART 13 — FUTURE TECHNOLOGY MIGRATION
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Document migration strategies for:
- databases
- storage
- APIs
- Android/platform
- schemas
- search engines
- map systems
- AI providers
- automation systems

Preserve identifiers and provenance during migrations.

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PART 14 — AI CONTINUITY
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Preserve where applicable:
- model/provider
- model version
- workflow/prompt version
- source inputs
- generated outputs
- human review
- final decision

AI-generated historical suggestions must remain distinguishable from authoritative records.

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PART 15 — GLOBAL GOVERNANCE
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Define governance for:
- federation membership
- trust levels
- institutional responsibilities
- data stewardship
- source ownership
- dispute resolution
- privacy
- licensing
- data sovereignty
- access control

Do not invent governance agreements.

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PART 16 — ARCHIVE TRUST MODEL
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If a trust model is implemented, clearly distinguish:
- verified archive node
- recognized institution
- unverified node
- restricted node
- suspended node

Trust status must not imply that every historical claim from a trusted source is automatically correct.

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PART 17 — FUTURE-PROOF API
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Evaluate:
- stable identifiers
- version negotiation
- backward compatibility
- API versioning
- deprecation policy
- migration endpoints
- provenance preservation

Do not break existing clients without a documented migration path.

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PART 18 — SECURITY OF THE FEDERATION
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Review:
- node authentication
- authorization
- mutual trust
- API credentials
- signing keys
- synchronization endpoints
- replay protection
- audit logging
- compromised-node handling

Apply least privilege.

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PART 19 — PRIVACY & DATA SOVEREIGNTY
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Review:
- cross-border data movement
- institutional restrictions
- private research
- sensitive information
- API exposure
- synchronization
- archival copies

Only replicate data that the source/license permits.

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PART 20 — FEDERATION MONITORING
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Monitor where appropriate:
- node availability
- synchronization status
- replication lag

- integrity status
- failed transfers
- conflicts
- archive health

Use actual measurements.

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PART 21 — FAILURE HANDLING
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Every federation workflow must define:

DETECT
→ CLASSIFY
→ STOP/ISOLATE
→ RECOVER
→ VERIFY
→ RESUME OR ESCALATE
→ AUDIT

Do not automatically trust a compromised or corrupted node.

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PART 22 — FINAL ARCHITECTURE AUDIT
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Review Phases 1–14 as one system.

Evaluate:
- security
- data integrity
- provenance
- privacy
- preservation
- scalability
- AI governance
- institutional governance
- portability
- disaster recovery
- federation readiness

Identify architectural contradictions and technical debt.

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PART 23 — LONG-TERM MAINTAINABILITY
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Evaluate:
- dependency lifecycle
- documentation quality
- migration burden
- operational complexity
- staffing/maintenance requirements
- monitoring burden
- archival maintenance

Avoid unnecessary complexity.

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PART 24 — DOCUMENTATION
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Create/update:

docs/FEDERATED-ARCHITECTURE.md
docs/ARCHIVE-NODES.md
docs/CROSS-ARCHIVE-INTEROPERABILITY.md
docs/ARCHIVE-INDEPENDENCE.md
docs/PORTABLE-ARCHIVAL-PACKAGE.md
docs/CRYPTOGRAPHIC-INTEGRITY.md
docs/FORMAT-PRESERVATION.md
docs/FEDERATED-PROVENANCE.md
docs/CONFLICT-RESOLUTION.md
docs/FEDERATED-SYNCHRONIZATION.md
docs/DISASTER-INDEPENDENCE.md
docs/FUTURE-MIGRATION.md
docs/AI-CONTINUITY.md
docs/GLOBAL-GOVERNANCE.md
docs/ARCHIVE-TRUST-MODEL.md
docs/FUTURE-PROOF-API.md
docs/FEDERATION-SECURITY.md
docs/DATA-SOVEREIGNTY.md
docs/FEDERATION-MONITORING.md
docs/FINAL-ARCHITECTURE-AUDIT.md
docs/PHASE-15-ROADMAP.md

Document actual implementation only.

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PHASE 15 ACCEPTANCE GATE
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[ ] Phase 14 baseline reviewed
[ ] Federated archive architecture
[ ] Archive node model
[ ] Cross-archive interoperability
[ ] Archive independence
[ ] Portable archival package
[ ] Cryptographic integrity
[ ] Long-term format strategy
[ ] Federated provenance
[ ] Conflict resolution
[ ] Global synchronization
[ ] Disaster independence
[ ] Future technology migration
[ ] AI continuity
[ ] Global governance
[ ] Archive trust model
[ ] Future-proof API
[ ] Federation security
[ ] Privacy/data sovereignty
[ ] Federation monitoring
[ ] Failure handling
[ ] Final architecture audit
[ ] Long-term maintainability
[ ] Documentation

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FINAL REPORT
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PHASE 15 STATUS: READY / NOT READY

For every implemented capability report:
- what was built
- evidence
- tests performed
- limitations
- security considerations
- privacy considerations
- provenance considerations
- operational impact

Use actual metrics only.

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ARCHITECTURE SCORECARD
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Federation readiness:
[actual assessment]

Archive independence:
[actual assessment]

Portability:
[actual assessment]

Integrity:
[actual assessment]

Provenance:
[actual assessment]

Disaster recovery:
[actual assessment]

Security:
[actual assessment]

Privacy/data sovereignty:
[actual assessment]

AI continuity:
[actual assessment]

Maintainability:
[actual assessment]

Do not invent numeric scores without a defined methodology.

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BLOCKERS
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List every issue preventing Phase 15 from being READY.

For each:
- severity
- component
- issue
- evidence
- impact
- remediation
- status

Never include secrets.

==
ROADMAP
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NOW:
[items]

NEXT:
[items]

LATER:
[items]

DO NOT BUILD YET:
[items]

==
POST-PHASE-15 DECISION
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Do NOT automatically create Phase 16.

After completing Phase 15, determine whether GraveAtlas has reached a stable long-term architecture.

If additional work is required, document the evidence-based reasons and specific unmet objectives rather than inventing another phase merely to extend the roadmap.

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STOP CONDITION
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Do NOT start Phase 16 automatically.

Do NOT add paid services without explicit approval.

Do NOT expose secrets.

Do NOT fabricate archive nodes, institutions, partnerships, data or results.

Do NOT silently modify authoritative historical records.

Do NOT publish restricted information.

Do NOT perform destructive bulk operations.

Do NOT perform uncontrolled production load testing.

STOP after the Phase 15 final report.

WAIT FOR EXPLICIT INSTRUCTIONS.